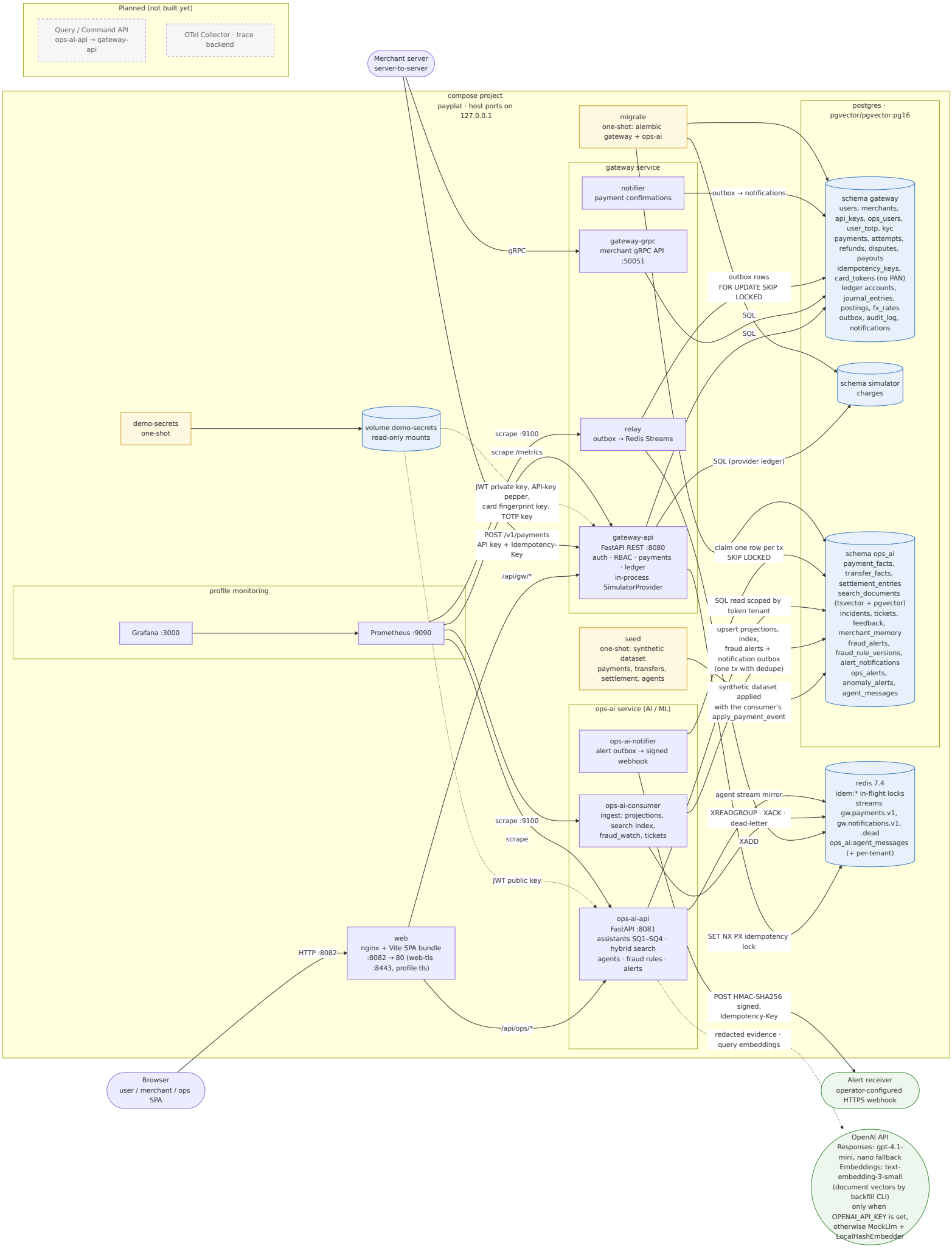

Planned (not built yet)
Query / Command API
ops-ai-api → gateway-api
OTel Collector · trace backend
Merchant server
server-to-server
compose project
payplat · host ports on 127.0.0.1
migrate
one-shot: alembic
gateway + ops-ai
postgres · pgvector/pgvector:pg16
gateway service
notifier
payment confirmations
outbox → notifications
schema gateway
users, merchants, api_keys, ops_users, user_totp, kyc
payments, attempts, refunds, disputes, payouts
idempotency_keys, card_tokens (no PAN)
ledger accounts, journal_entries, postings, fx_rates
outbox, audit_log, notifications
gRPC
gateway-grpc
merchant gRPC API
:50051
outbox rows
FOR UPDATE SKIP LOCKED
SQL
SQL
schema simulator
charges
demo-secrets
one-shot
volume demo-secrets
read-only mounts
scrape :9100
scrape /metrics
relay
outbox → Redis Streams
SQL (provider ledger)
JWT private key, API-key pepper,
card fingerprint key,
TOTP key
POST /v1/payments
API key + Idempotency-Key
gateway-api
FastAPI REST :8080
auth · RBAC · payments · ledger
in-process SimulatorProvider
claim one row per tx
SKIP LOCKED
schema ops_ai
payment_facts, transfer_facts, settlement_entries
search_documents (tsvector + pgvector)
incidents, tickets, feedback, merchant_memory
fraud_alerts, fraud_rule_versions, alert_notifications
ops_alerts, anomaly_alerts, agent_messages
/api/gw/*
profile monitoring
SQL read scoped by token tenant
Grafana :3000
Prometheus :9090
seed
one-shot: synthetic dataset
payments, transfers, settlement, agents
upsert projections, index,
fraud alerts + notification outbox
(one tx with dedupe)
synthetic dataset applied
with the consumer's apply_payment_event
ops-ai service (AI / ML)
ops-ai-notifier
alert outbox → signed webhook
redis 7.4
idem:* in-flight locks
streams
gw.payments.v1, gw.notifications.v1, .dead
ops_ai:agent_messages (+ per-tenant)
agent stream mirror
XREADGROUP · XACK · dead-letter
XADD
ops-ai-consumer
ingest: projections, search index, fraud_watch, tickets
scrape :9100
scrape
JWT public key
ops-ai-api
FastAPI :8081
assistants SQ1–SQ4 · hybrid search
agents · fraud rules · alerts
SET NX PX idempotency lock
web
nginx + Vite SPA bundle
:8082 → 80 (web-tls
:8443, profile tls)
HTTP :8082
POST HMAC-SHA256 signed, Idempotency-Key
/api/ops/*
redacted evidence · query embeddings
Browser
user / merchant / ops SPA
Alert receiver
operator-configured HTTPS webhook
OpenAI API
Responses: gpt-4.1-mini, nano fallback
Embeddings: text-embedding-3-small (document vectors by backfill CLI)
only when OPENAI_API_KEY is set, otherwise MockLlm + LocalHashEmbedder